LOS_LAB ASSIGNMENT - 3

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Grep, Regex and File Operators

Part A — Basic grep

Exercise 1 — Search File Contents

```
grep "wget" payload.sh
grep "nc" payload.sh
grep "chmod" payload.sh
grep -i "WGET" payload.sh
grep -n "/tmp" payload.sh
grep -c "/tmp" payload.sh
grep -o "wget" payload.sh
grep -w "nc" payload.sh
```

Exercise 2 — Compare Files

```
grep "wget" payload.sh update.sh cleanup.sh

grep "/tmp" payload.sh update.sh cleanup.sh
```

Which file contains wget? payload.sh (assuming the search matches here).

Which files do not contain wget? update.sh and cleanup.sh.

Which files contain /tmp? payload.sh (or any file printed in the terminal prefix).
Use grep with multiple filenames: The commands above demonstrate passing multiple files as sequential arguments.

Exercise 3 — Inverse Search

```
grep -v "INFO" system.log
grep -v "ERROR" system.log
grep -v "WARNING" system.log
grep -v -c "INFO" system.log
```

Part B — Security Log Analysis

Exercise 4 — Analyze system.log

```
grep "ERROR" system.log
grep "WARNING" system.log
grep "FAILED" system.log
grep -c "ERROR" system.log
grep -n "WARNING" system.log
grep -i "error" system.log
grep -q "Malware" system.log
```

Exercise 5 — Context Investigation

```
grep -A 2 "ERROR" system.log
grep -B 2 "ERROR" system.log
grep -C 1 "ERROR" system.log
grep -C 2 "FAILED" system.log
```

Why can examining events before and after an error be useful during incident investigation?
Before: Reveals the root cause, user actions, or preliminary exploitation attempts leading up to the crash or alert.
After: Exposes subsequent fallout, secondary payloads executed, error propagation, or system damage following the initial failure.

Part C — egrep / grep -E

Exercise 6 — Search Multiple Security Events

```
grep -E "ERROR|WARNING|FAILED" system.log
```

Exercise 7 — Multiple Suspicious Commands

```
grep -E "wget|curl|nc|chmod" payload.sh
grep -E -c "wget|curl|nc|chmod" payload.sh
```

Which patterns were found? wget, nc, chmod (based on Exercise 1 contents).

Which pattern was not found? curl.

How many lines contain at least one of these patterns? Use the second command above.

Exercise 8 — Search Multiple Files

```
grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
grep -lE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
grep -nE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
grep -cE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
```

Part D — patterns.txt

Exercise 9 — Pattern File Investigation

```
grep -f patterns.txt payload.sh
grep -f patterns.txt system.log
grep -f patterns.txt config.conf
grep -f patterns.txt *.sh
grep -l -f patterns.txt *
grep -h -f patterns.txt *
```

Which files contain indicators listed in patterns.txt? The output generated by grep -l -f patterns.txt * directly lists these files.

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

```
grep "^malware" filelist.txt
grep "^script" filelist.txt
grep "\.sh$" filelist.txt
grep "\.py$" filelist.txt
grep "\.log$" filelist.txt
grep "^p.*\.sh$" filelist.txt
```

Exercise 11 — Character Classes

```
grep -E "malware[12]\.sh" filelist.txt
grep "[0-9]" filelist.txt
grep "[12]" filelist.txt
grep "^[mp]" filelist.txt
grep "^[su]" filelist.txt
```

Exercise 12 — Negated Character Classes

```
grep "^[^m]" filelist.txt
grep "^[^mp]" filelist.txt
grep -E "malware[^0-9]" filelist.txt
```

Part F — File Operators

Exercise 13 — File Existence

```
#!/bin/bash
filename=$1
if [ -e "$filename" ]; then
   echo "File exists"
else
   echo "File does not exist"
fi
```

Test execution: ./script.sh payload.sh, ./script.sh report.txt, ./script.sh unknown.sh

Exercise 14 — File Type

```
#!/bin/bash
path=$1
if [ -f "$path" ]; then
   echo "Regular file"
elif [ -d "$path" ]; then
   echo "Directory"
else
   echo "Neither"
fi
```

Test execution: Run against payload.sh, system.log, quarantine, backup, and unknown.

Exercise 15 — File Permissions

```
#!/bin/bash
file=$1
echo "File: $file"
echo ""
[ -r "$file" ] && r="Yes" || r="No"
[ -w "$file" ] && w="Yes" || w="No"
[ -x "$file" ] && x="Yes" || x="No"
echo "Readable   : $r"
echo "Writable   : $w"
echo "Executable : $x"
```

Exercise 16 — File Data Check

```
#!/bin/bash
file=$1
if [ -s "$file" ]; then
   echo "File contains data"
else
   echo "File is empty"
fi
```

Part G

Exercise 17 — File Validation Using [ ]

```
#!/bin/bash
file=$1
if [ -e "$file" ] && [ -f "$file" ] && [ -r "$file" ] && [ -s "$file" ]; then
   echo "Valid file: Exists, regular, readable, and non-empty."
else
   echo "Invalid file or fails criteria."
fi
```

Exercise 18 — [ ] + grep

```
#!/bin/bash
file=$1
if [ -e "$file" ] && [ -f "$file" ] && [ -r "$file" ] && [ -s "$file" ]; then
   if grep -q "wget" "$file"; then
      echo "Selected indicator detected – investigate further"
   else
      echo "Selected indicator not detected"
   fi
else
   echo "File checks failed."
fi
```

Part H

Exercise 19 — Repeat File Tests Using [[ ]]

```
#!/bin/bash
file=$1
if [[ -e "$file" && -f "$file" && -r "$file" && -s "$file" ]]; then
   echo "Valid file: Exists, regular, readable, and non-empty."
else
   echo "Invalid file or fails criteria."
fi
```

Exercise 20 — Regex Using [[ ]]

```
#!/bin/bash
filename=$1
if [[ "$filename" =~ \.sh$ ]]; then
  echo "Shell script identified for inspection"
else
  echo "File does not have a .sh extension"
fi
```

Exercise 21 — Multiple Extensions Using [[ ]]

```
#!/bin/bash
filename=$1
if [[ "$filename" =~ \.(sh|py|pl)$ ]]; then
  echo "Script file – inspect if required"
fi
```

Exercise 22 — Executable File Content Analysis

```
#!/bin/bash
FILE="$1"

if [ -e "$FILE" ] && [ -f "$FILE" ] && [ -x "$FILE" ]; then
  grep "wget" "$FILE"
fi
```

Test with: payload.sh, update.sh, report.txt

Exercise 23 — egrep + File Operators

```
#!/bin/bash
FILE="$1"

if [ -f "$FILE" ] && [ -r "$FILE" ] && [ -s "$FILE" ]; then
  if grep -E "wget|curl|nc|chmod" "$FILE" > /dev/null 2>&1; then
    echo "One or more selected indicators detected"
  else
    echo "Selected indicators not detected"
  fi
fi
```

Exercise 24 — Sensitive Information Search

```
#!/bin/bash
FILE="config.conf"

if [ -e "$FILE" ] && [ -f "$FILE" ] && [ -r "$FILE" ] && [ -s "$FILE" ]; then
   grep -E "password|username" "$FILE"
fi
```

Why should configuration files containing authentication information receive additional protection?

Configuration files may contain usernames, passwords, API keys, or other sensitive authentication information. Unauthorized access can allow attackers to gain access to systems or services.

Part K

Exercise 25 — Suspicious File Triage (file_check.sh)

```
#!/bin/bash

read -p "Enter filename: " FILE
echo -e "\n----- FILE INVESTIGATION -----\n"

check() { [ "$1" ] && echo "$2: Yes" || echo "$2: No"; }

if [ ! -e "$FILE" ]; then echo "Exists       : No"; exit 1; fi

check "-e $FILE" "Exists       "
check "-f $FILE" "Regular File "
check "-d $FILE" "Directory    "
check "-r $FILE" "Readable     "
check "-w $FILE" "Writable     "
check "-x $FILE" "Executable   "
check "-s $FILE" "Non-empty    "

[[ "$FILE" =~ \.sh$ ]] && echo "Shell Script  : Yes" || echo "Shell Script  : No"
echo ""
```

```
if [ -r "$FILE" ] && [ -s "$FILE" ]; then
    grep "wget" "$FILE" > /dev/null 2>&1
    check "[ $? -eq 0 ]" "wget        "

    grep -E "wget|curl|nc|chmod" "$FILE" > /dev/null 2>&1
    if [ $? -eq 0 ]; then
        echo "Selected indicators : Found"
        echo -e "\nFurther inspection recommended"
    else
        echo "Selected indicators : Not Found"
    fi
fi
```

Example test with payload.sh:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists : Yes

Regular File : Yes

Directory : No

Readable : Yes

Writable : Yes

Executable : Yes

Non-empty : Yes

Shell Script : Yes

wget : Found

Selected indicators : Found

Further inspection recommended